Mating Connector:
Housing: Molex 0436450200
Precrimped 18AWG wire 450mm:
Black: Molex 2147601225
Red: Molex 2147602225

Reverse Polarity Protection

Q14
CSD19534Q5A

J10
POWER
V_IN
V_IN_RTN
Molex
0436500200

D1
SMDJ24CA
24V

C64
10uF-50V-10%-X7R-1210

C62
0.1uF-50V-10%-0603

U11
LM74700QDBVRQ1
VCAP
EN
GND
A
G
C

C55
10uF-50V-10%-X7R-1210

R29
5.1k-1%-0603

R31
5.1k-1%-0603

R25
5.1k-1%-0603

D2
PWR

VIN_PS

File: buck-boost-power-supply.kicad_sch
2/5
+V_IN
-V_IN
+V_OUT
-V_OUT
15V 3A Buck-Boost Power Supply

Power Supply Back
Feed Protection

Q1
CSD19534Q5A

C1
10uF-50V-10%-X7R-1210

C15
0.1uF-50V-10%-0603

U2
LM74700QDBVRQ1
VCAP
EN
GND
A
G
C

C16
10uF-50V-10%-X7R-1210

J4
V_OUT
V_OUT_RTN
15V
Molex
0436500200

Mating Connector:
Housing: Molex 0436450200
Precrimped 18AWG wire 450mm:
Black: Molex 2147601225
Red: Molex 2147602225

Mating Connector:
Housing: Molex 5040510401
Precrimped 24AWG wire 150mm: Molex 0797581003

J7
USB1
Molex
5040500491

File: canable2.kicad_sch
3/5
USB_DP
USB_DM
3V3_USB
GND_CAN
CANable2.0
CAN_HC
CAN_LC

J9
CAN1
MFG P/N: 0705550036
MFG: Molex

J1
Terminal

J2
Terminal
Keystone Electronics
7774

File: NX-Button-Interface.kicad_sch
4/5
NX-Button Interface

File: usb-serial.kicad_sch
5/5
+5V J401
3V3 J401
RXD
TXD
RESET
FORCE_RECOVERY
USB Serial

VDD_5V_SYS
3V3_BTN
UART2_TXD
UART2_RXD
SYS_RESET
REC
VDD_5V_SYS
NX_PC_LED-

PH1RB-15-UA
Adam Tech
15 3V3_BTN
14
13
12 NX_PC_LED-
11 VDD_5V_SYS
10 UART2_RXD
9 UART2_TXD
8 AC OK
7 Auto Power On Disable
6
5 SYS_RESET
4
3 REC
2 GND
1 SLEEP/WAKE*
NX CONTROL & UART
J3
GND1

D3
PC

SW2
RECOVERY

SW1
RESET
GND1

Light Pipe
LP1
Bivar
Bivar-SLP3-150-100-F

Light Pipe
LP2
Bivar
Bivar-SLP3-150-100-F

Light Pipe
LP3
Bivar
Bivar-SLP3-150-100-F

Light Pipe
LP4
Bivar
Bivar-SLP3-150-100-F

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H3 MountingHole
H10 MountingHole
H11 MountingHole
H4 MountingHole
H5 MountingHole
H7 MountingHole
H8 MountingHole

H1 1.152mm ToolingHole
H12 1.152mm ToolingHole
H2 1.152mm ToolingHole
H13 1.152mm ToolingHole

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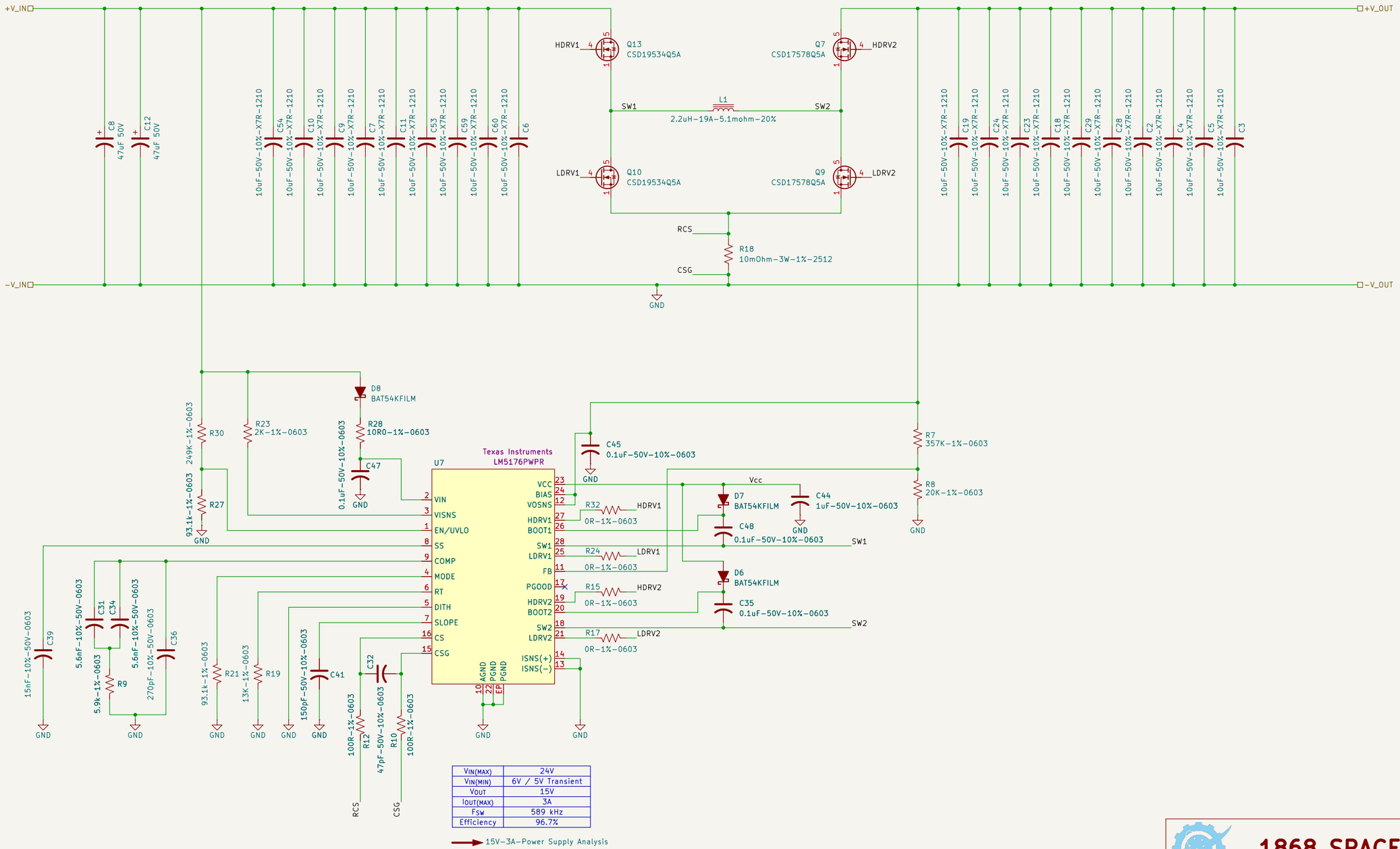
open source hardware

1868 SPACE COOKIES

Sheet: /
File: TVS-Adapter-v1.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-31	Rev: 1
KiCad E.D.A. 10.0.8		Id: 1/5



VIN(MAX)	24V
VIN(MIN)	6V / 5V Transient
VOUT	15V
IOUT(MAX)	3A
FSW	589 kHz
Efficiency	96.7%

15V−3A−Power Supply Analysis

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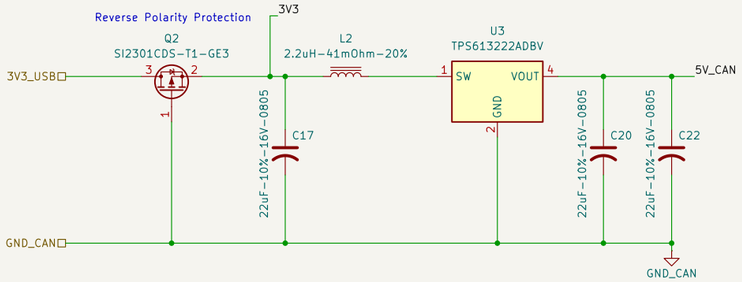
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File: buck−boost−power−supply.kicad_sch

Title: The Vision System Adapter

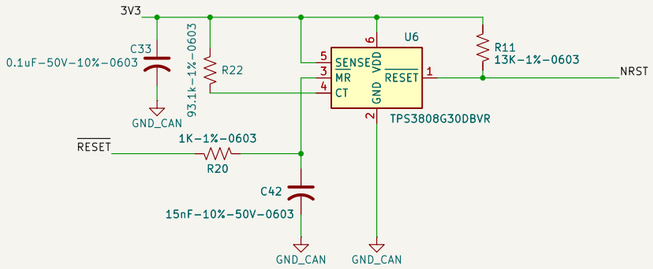
Size: B	Date: 2026−10−31	**Rev: 1**
KiCad E.D.A. 10.0.8		Id: 2/5

5V-250mA-Power Supply Analysis

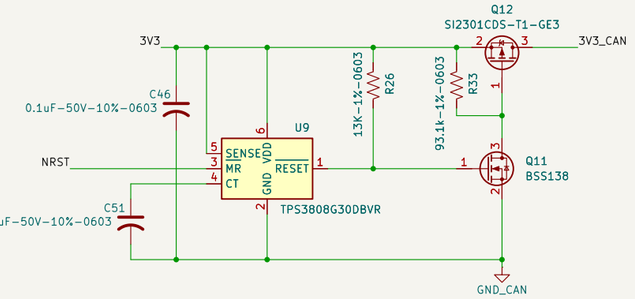
Reverse Polarity Protection

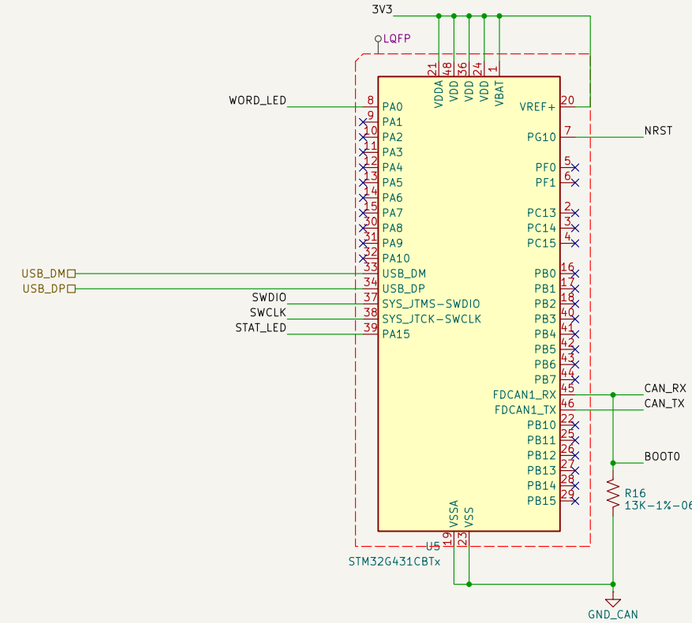


POWER ON RESET
300ms

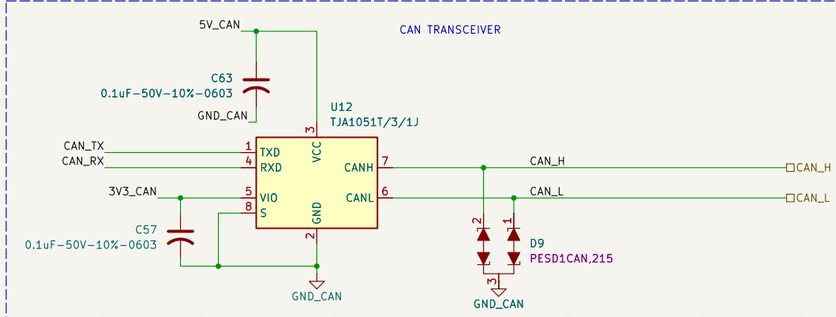


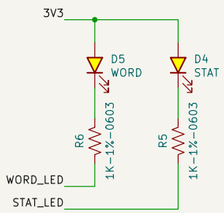
CAN BOOTO POWER DELAY
0.5 seconds

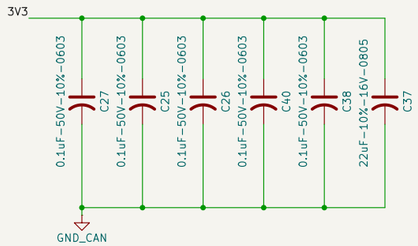




CAN TRANSCEIVER







DEBUG HEADER



NOTES:
- CANABLE circuit interfaces with the M.2 Key-E adapter plugged into the J401. Input power must be no greater than 3.6V

CANable 2.0 Schematic

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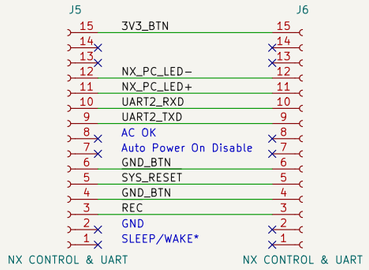
open source hardware

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Sheet: /CANable2.0/
File: canable2.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-31	**Rev: 1**
KiCad E.D.A. 10.0.8		Id: 3/5



H6
M3.7 MountingHole

H9
M3.7 MountingHole

SPACE COOKIES LOGO

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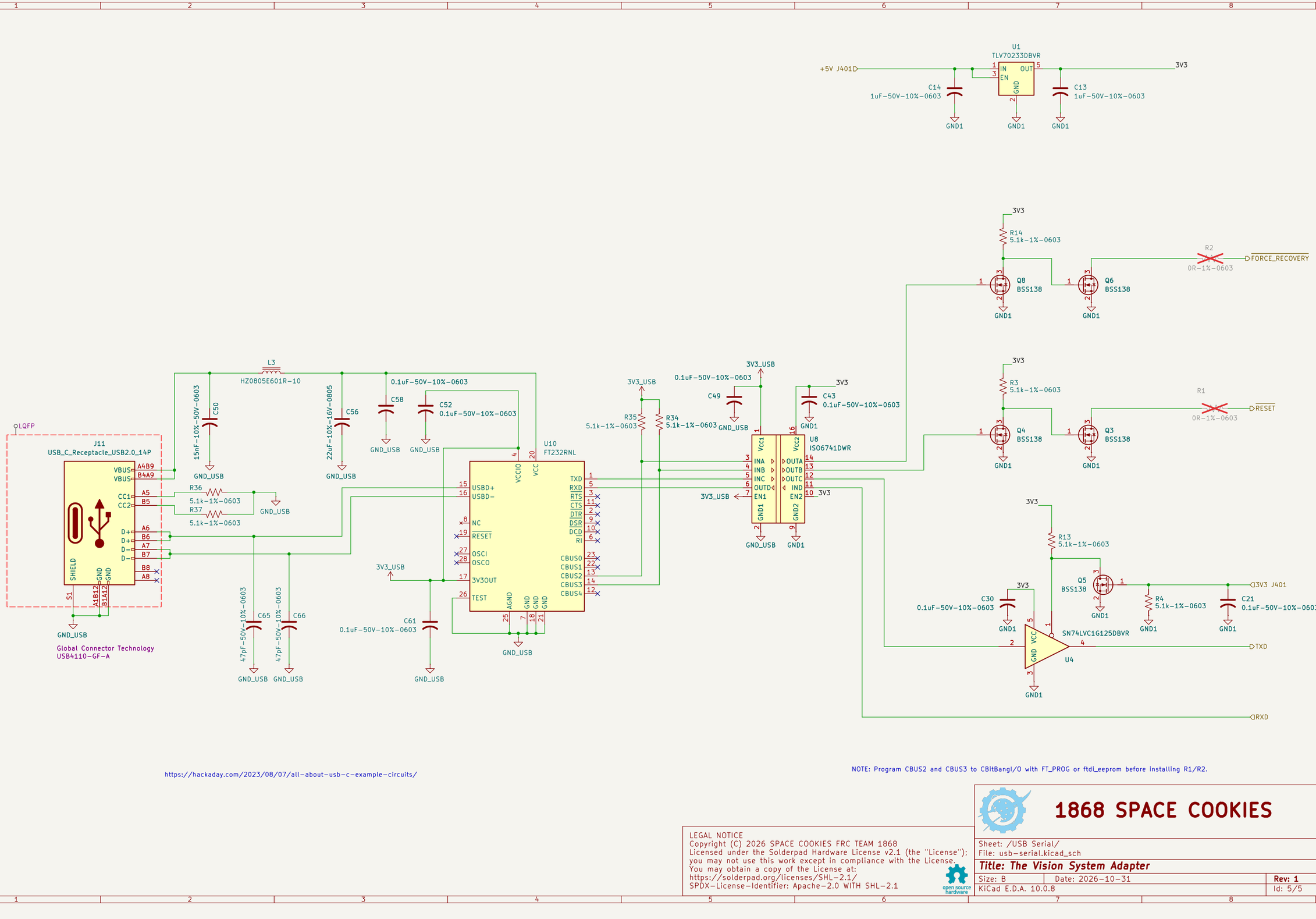
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Sheet: /NX-Button Interface/
File: NX-Button-Interface.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-31	**Rev: 1**
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Sheet: /USB Serial/
File: usb-serial.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-31	**Rev: 1**
KiCad E.D.A. 10.0.8		Id: 5/5